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**ELEC2151
CIRCUIT THEORY
PROJECT REPORT**

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1. ABSTRACT

This baby monitoring system using Internet of Things (IoT) technology was developed to help parents track their infant's environment and well-being. The system measures room temperature and humidity while also detecting crying sounds. All data is sent to a web platform through Wi-Fi, allowing parents to check conditions remotely. When crying is detected, the system immediately notifies parents through the web application. The system includes data logging capabilities, enabling parents to export historical data in both PDF and CSV formats for detailed analysis and record-keeping. This feature allows for long-term tracking of environmental conditions and crying patterns, which can be valuable for understanding infant behavior and maintaining optimal conditions. This monitor helps maintain a safe environment for babies and lets parents respond quickly when needed. The system provides a comprehensive and effective solution for modern infant monitoring needs, combining real-time alerts with data analysis capabilities.

2. INTRODUCTION

In recent years, the use of Internet of Things (IoT) technology in childcare has changed how parents monitor their babies' well-being. Traditional baby monitors, which mainly focused on transmitting audio, have developed into advanced systems that can provide detailed environmental monitoring and data analysis. This improvement meets the growing need for better and more accessible information about infant care.

The project uses IoT principles, providing wireless communication via web-server and web-based interfaces to create a solution for parents. It monitors important environmental factors like temperature and humidity in real-time while also detecting if the baby is crying. Bringing these features together, the system provides more detailed information about the baby's environment compared to older baby monitor designs.

One of the main challenges in baby care is keeping the environment safe while quickly responding to the baby's needs. This system solves this by sending instant alerts when the baby cries and allowing parents to check environmental conditions remotely. Additionally, it includes data logging and export options in PDF and CSV formats. This helps parents track changes over time and find patterns that may affect their baby's comfort and health.

3. STRUCTURE

This section will discuss the hardware and software used in the project.

3.1 Sensors, Modules and Control Board

1) DHT11 Temperature and Humidity Sensor

The DHT11 temperature sensor compares the ambient temperature and humidity using its internal electrodes and produces analog data. Studies suggest that the temperature in an environment where a baby is present should be maintained between 20 to 30 degrees Celsius, and the humidity level should be kept between 30% and 65%.

2) KY-038 Sound Sensor

The sound sensor detects sound waves in the environment and converts them into a magnitude. With the KY-038 sound sensor, if no noise is detected in the environment, the average sound level measured over 10 seconds is sent to the interface. If noise is detected, the highest detected sound level is transmitted to the interface. When the sound level is 55 or above, the system shows alert to parents.

3) ESP8266-01S Wi-Fi Module

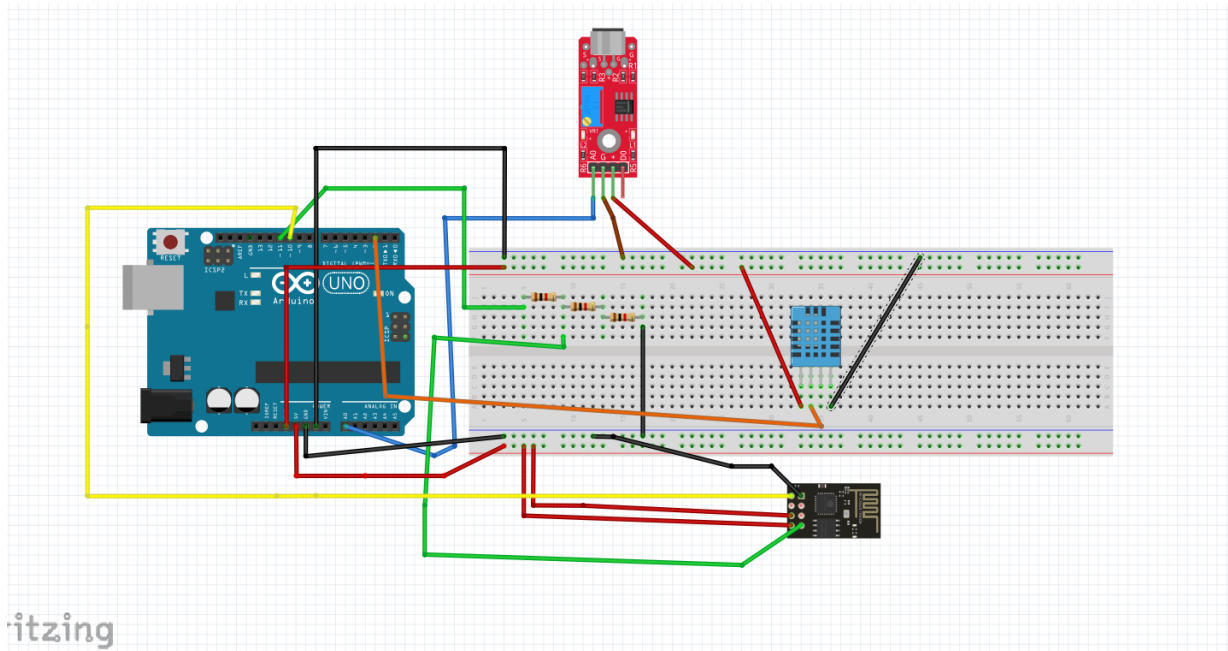
The ESP8266-01S module receives data from the sensors and sends it to the server's database, where the web interface is hosted, by making an HTTP POST request to add the information.

4) Arduino Uno

Arduino is an open-source hardware and software platform, and the Arduino Uno control board used in this project enables communication between sensors and modules.

3.2 Electrical Design

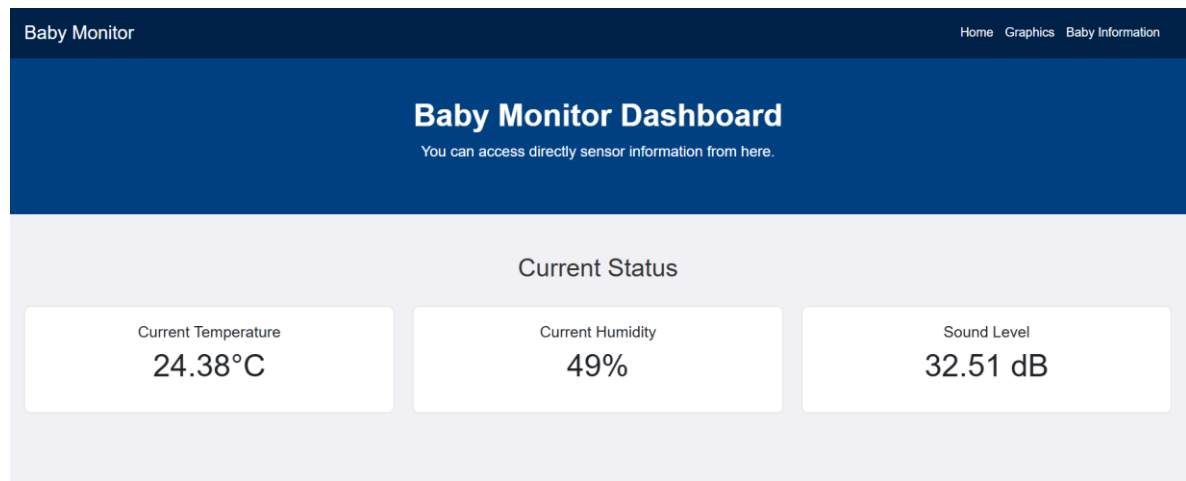
For the connection of the electronic circuit, apart from the above sensors, 16V 100 μ F capacitor, 3 1K ohm resistors and male-male, male-female jumper cables were used for their connections. The full design is shown in the diagram below:



Pic.1: Circuit diagram

3.3 Web Page

The ESP8266-01S module sends the data read from the sensor to the database (*baby_records*) via an HTTP Post Request. The website, which is used as the interface, employs JavaScript and PHP in the backend. Through API management, the data stored in the database is processed by the interface. The website has features such as displaying warnings if anomalies are detected in the recorded data, plotting the last 30 data points on a graph, displaying baby information, and downloading data between selected dates in PDF or CSV format.



Pic.2: Shows the general information



Pic.3: Graph-drawing feature of web-site

The 'Baby Information' section displays a white box with the following text: 'İsim: Test Baby', 'Doğum Tarihi: 2025-01-01', and 'Açıklamalar: Hello World!'. Below this, the 'Download Data' section includes a form with 'Start date: gg.aa.yyyy', 'End date: gg.aa.yyyy', 'Format: CSV', and a green 'Download' button.

Pic.4: Baby Information and downloading data feature

4. CONCLUSION

This baby monitoring system, designed using Internet of Things (IoT) technology, provides a practical and effective solution for parents to monitor their babies. The system continuously checks important factors like room temperature, humidity, and the baby's crying. By using sensors and wireless communication, it ensures that parents can easily track their baby's environment in real-time from anywhere.

One of the main strengths of this system is its ability to alert parents as soon as the baby starts crying. This feature is crucial as it allows parents to respond quickly to their baby's needs, especially when they are not in the same room. Another strength is the monitoring of environmental conditions such as temperature and humidity. Keeping the room within the right temperature (20-30°C) and humidity (30%-65%) is very important for a baby's health and comfort, and this system helps parents maintain those conditions easily.

In addition to real-time monitoring, the system also has a data logging feature. This allows parents to store and review historical data, which can be downloaded in PDF or CSV format. By doing so, they can analyze changes in the baby's environment over time, looking for patterns or irregularities that could affect their baby's comfort or health. This long-term tracking is especially useful for understanding how different factors might influence the baby's well-being.

The system's user interface, accessible via a web page, is designed to be simple and easy to use. It allows parents to see current data, check historical trends, and receive notifications when something unusual happens. This interface makes it easy to track everything from anywhere, giving parents peace of mind that they can always stay informed about their baby's condition.

Overall, this system offers several benefits for modern baby care. It helps parents keep their baby safe and comfortable by providing instant alerts when needed, offering remote monitoring of the baby's environment, and giving parents the tools to analyze past data. The combination of real-time monitoring and long-term tracking, along with the ease of use provided by the web interface, makes this system a valuable tool for parents. It ensures a safer and more comfortable environment for babies while also helping parents respond quickly and effectively to their baby's needs.

5. REFERENCES

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